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Developing Polypropylene Covalent Adaptable Networks with Complete Cross-Link Density Recovery after Reprocessing by Free-Radical Reactive Processing with Vinyl Aromatic Additives

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Abstract

Free-radical reactive processing of thermoplastic polyethylene or ethylene-containing copolymers with a dynamic covalent cross-linker enables the synthesis of covalent adaptable networks (CANs). While effective for polyethylene, this approach is hindered in polypropylene (PP) due to the propensity of tertiary carbon radicals in PP to undergo β -scission during grafting. We have developed PP-based CANs via one-step, radical-based reactive processing using dicumyl peroxide (a radical initiator), bis(4-methacryloyloxyphenyl) disulfide (BPMA, an aromatic disulfide-based dynamic covalent cross-linker), and, to stabilize the radicals and promote cross-linking, vinyl aromatic additives. Adding 2-vinylnaphthalene (VN) at 2.0 or 4.0 molar equivalents to BPMA effectively suppressed β -scission in PP and enabled robust CAN formation. Divinylbenzene (DVB) at 0.5 molar equivalent to BPMA also enabled PP CAN formation, but due to its additional function as a permanent cross-linker, further increases in DVB level led to the percolation of permanent cross-links and loss of reprocessability. Relative to PP, all PP CANs exhibited significant melt-state creep suppression; the best creep resistance (and highest cross-link density) exhibited by a PP CAN, a factor of 40 better than that of PP, was prepared using a combination of 2.0 molar equivalents of VN and 0.5 molar equivalent of DVB relative to BPMA. Notably, each PP CAN exhibited complete recovery of cross-link density after two reprocessing cycles. Thus, this study represents a successful, one-step, additive-based approach for making robust and reprocessable PP CANs.

Keywords

covalent adaptable networks, polypropylene, aromatic additives, reactive processing, free radical, cross-link density

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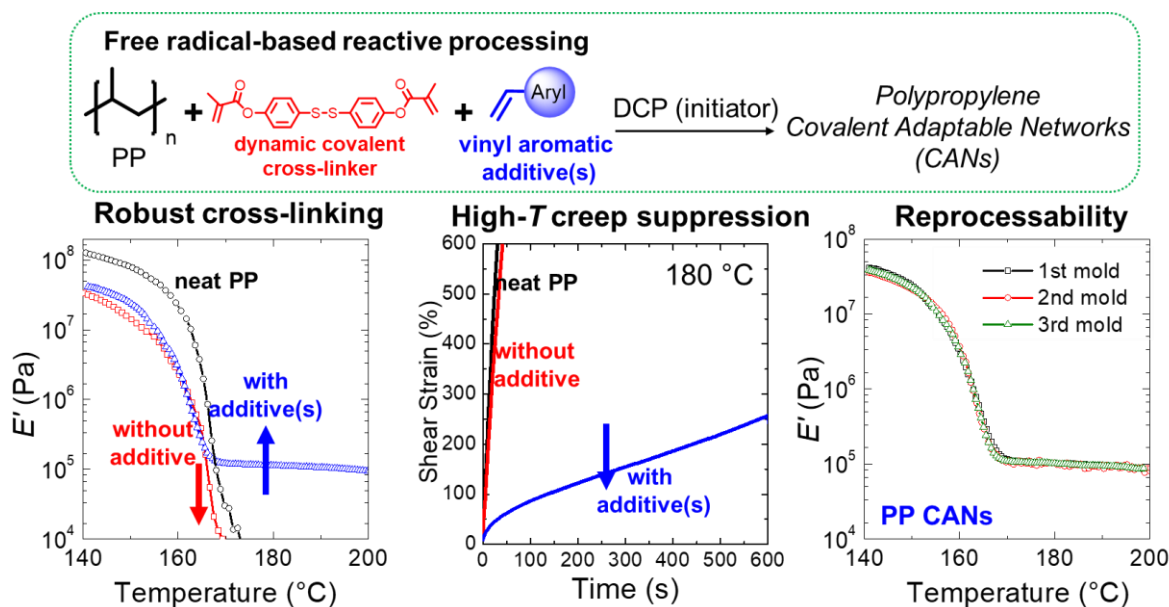
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during grafting. We have developed PP-based CANs via one-step, radical-based reactive processing using dicumyl peroxide (a radical initiator), bis(4-methacryloyloxyphenyl) disulfide (BPMA, an aromatic disulfide-based dynamic covalent cross-linker), and, to stabilize the radicals and promote cross-linking, vinyl aromatic additives. Adding 2-vinylnaphthalene (VN) at 2.0 or 4.0 molar equivalents to BPMA effectively suppressed β -scission in PP and enabled robust CAN formation. Divinylbenzene (DVB) at 0.5 molar equivalent to BPMA also enabled PP CAN formation, but due to its additional function as a permanent cross-linker, further increases in DVB level led to the percolation of permanent cross-links and loss of reprocessability. Relative to PP, all PP CANs exhibited significant melt-state creep suppression; the best creep resistance (and highest cross-link density) exhibited by a PP CAN, a factor of 40 better than that of PP, was prepared using a combination of 2.0 molar equivalents of VN and 0.5 molar equivalent of DVB relative to BPMA. Notably, each PP CAN exhibited complete recovery of cross-link density after two reprocessing cycles. Thus, this study represents a successful, one-step, additive-based approach for making robust and reprocessable PP CANs.

INTRODUCTION

Polyolefins (POs), a class of synthetic polymers that includes polyethylene (PE) and polypropylene (PP), represent the majority of global plastic production; PE and PP are the most widely manufactured plastics, comprising approximately 36% and 21% of global plastic production by weight, respectively.^{1,2} These polymers are valued for their mechanical properties, low cost, and ease of processing, making them suitable for various applications such as containers, films, and packaging.³ Despite their large-scale production, fewer than 10% of waste POs are recycled.⁴ Most discarded POs are incinerated to generate energy or sent to landfills, resulting in significant environmental issues.⁴ Traditional recycling methods for thermoplastic POs, such as re-extrusion or remolding, can lead to quality degradation due to the high temperatures needed for melt processing.⁵ Consequently, this type of recycling often results in downcycling, where the materials are repurposed for lower-value uses.^{6,7}

To promote recycling, the industrial and scientific communities have explored higher-value applications for thermoplastic POs, including both virgin and recycled materials.^{8,9} Cross-linked polyethylene (XLPE), for example, is produced by the recombination of PE macroradicals generated through radical-based reactive processing or irradiation.¹⁰ Compared to thermoplastic PE, XLPE shows improved thermal stability, impact strength, and chemical resistance, making it suitable for high-value uses such as pipe systems, foam, and cable insulation.^{11,12} However, a major limitation of XLPE is its inability to be mechanically recycled due to the permanent covalent cross-links that prevent melt flow.¹³

Efforts to address the recyclability challenge after cross-linking have led to advances in upcycling POs or PO blends into reprocessable covalent adaptable networks (CANs).¹⁴⁻³³ CANs incorporate dynamic covalent cross-links that facilitate structural reconfiguration under specific conditions (e.g. exposure to heat or light), thereby enabling recyclability for these cross-linked materials.^{34,35} Two types of chemistries allow dynamic covalent cross-links to undergo such reconfiguration. First, dissociative dynamic chemistries, including the Diels-Alder reaction,³⁶ hindered urea exchange,^{37,38} alkoxyamine chemistry,^{39,40} and dialkylamino disulfide chemistry,^{41,42} involve the reversible breakage of dynamic covalent cross-links, leaving two functional groups or radicals where there was previously one unit. Second, associative dynamic chemistries (networks incorporating these are sometimes called vitrimers^{15,16}) involve the simultaneous breaking and forming of bonds.⁴³ Examples include transesterification^{44,45}, transcarbamoylation,^{46,47} transamination,^{48,49} and boronic transesterification.⁵⁰ Additionally, some chemistries display both associative and dissociative mechanisms; examples include urethanes,^{51,52} thiourethanes,⁵³ hydroxyurethanes,^{46,54} and non-isocyanate thiourethanes.⁵⁵

While there are multiple methods for incorporating such dynamic chemistries into CANs, a recently developed approach offers a simple, drop-in modification of a commercial method for producing XLPE. This approach involves adding dimethacrylate-based dynamic covalent cross-linkers (featuring dialkylamino or aromatic disulfide linkages) alongside a peroxide during radical-based reactive processing of thermoplastic PE.^{19-21,32,33} Upon the decomposition of peroxide,

primary free radicals (RO*) abstract hydrogen atoms from PE, forming PE macroradicals that can graft onto the carbon-carbon double bonds on each end of the dynamic cross-linker or recombine with each other.^{10,19-21,32,33,56} In the former case, a dynamic cross-link is formed; in the latter, a permanent cross-link is formed. By keeping permanent cross-links below the percolation threshold while ensuring percolation of dynamic covalent cross-links, the CANs are reprocessable.⁵⁷

As with XLPE, cross-linking in PP also enhances important properties.⁵⁸ A low degree of cross-linking can improve PP's melt strength, preventing cell wall collapse in foaming applications;⁵⁹⁻⁶¹ a higher degree of cross-linking can improve the creep resistance, stress cracking behavior, and impact resistance of PP.⁶²⁻⁶⁴ However, cross-linked PP is not widely used due to challenges associated with conventional cross-linking methods.⁶⁵ In the conventional peroxide cross-linking approach, free radicals generated from the peroxide preferentially react with hydrogen atoms on the tertiary carbon of PP atoms.⁶⁶ This preference leads to the formation of tertiary macroradicals, which are highly susceptible to chain scission (β -scission).⁵⁸ Consequently, radical-based grafting and cross-linking approaches solely utilizing peroxide prove to be ineffective.⁵⁸ Given the challenges mentioned above, integrating dynamic bonds into PP-based polymers through radical-based reactive processing remains underexplored.

Indeed, most studies of PP CANs use a different approach. Commercially produced pre-functionalized PP-g-MA (made by reactive processing of PP homopolymer with maleic anhydride (MA), resulting in significant β -scission⁶⁷) has recently been repurposed as the major component of a polyurethane CAN by functionalization with hydroxyl groups followed by reaction with diisocyanates.⁶⁸ PP-g-MA can also be cross-linked using epoxy or thiol chemistry, affording ester^{69,70} or thioester^{71,72} units that are dynamic with the appropriate catalysis. However, one study⁶⁹ reported the absence of a rubbery plateau modulus (which, when present, reflects the network's percolated cross-link density according to Flory's ideal rubber elasticity theory⁷³) and sub-0.5 s stress relaxation times when above the melt temperature; both characteristics are indicative of thermoplastics rather than networks. Another study⁷² found that using very high MA levels (20–40 wt%) in PP-g-MA synthesis led to reasonable rubbery plateau tensile moduli (~ 0.1

MPa) and gel contents (44–55%); the authors noted that multiple molding cycles had little effect on material properties but did not report on the characterization of the remolded samples. A separate study⁷⁰ showed that PP-g-MA with 6 wt% MA could form CANs with moderate rubbery tensile storage moduli (0.44 MPa) and 42–58% gel fractions, though reprocessing led to reduced cross-link density and variations in the apparent melting temperature. A fourth study⁷¹ blended 15 wt% of a PP-g-MA CAN made from a brittle, low molecular weight (MW) precursor with 85 wt% high-MW thermoplastic PP; the resulting blend exhibited properties like the high-MW PP while outperforming blends containing 15 wt% uncross-linked low-MW PP-g-MA.

Efforts to cross-link PP without converting it into PP-g-MA are rare but not unprecedented. A 2024 study conducted peroxide-initiated vulcanization on a PP impact copolymer (containing a small level of ethylene repeat units for improved toughness) to create sulfide-based dynamic linkages in a single step.⁷⁴ However, attempts to measure the rubbery plateau moduli and gel contents of the resulting materials showed no signs of percolated cross-links.⁷⁴ Additionally, tensile tests of reprocessed samples did not demonstrate a full restoration of properties within experimental uncertainty, an essential characteristic when verifying CAN reprocessability.⁷⁴ Notably, a direct, single-step approach for forming percolated dynamic covalent cross-links in PP homopolymer, which is more challenging than in PP copolymer or PP-g-MA, remained unexplored until a very recent study by our group⁷⁵ that was inspired by earlier studies of PP grafting.^{76–82} These earlier studies demonstrated that styrene, used as an additive, can suppress β -scission by converting PP tertiary carbon radicals into stabilized carbon radicals.⁷⁶ Studies that grafted monomers like MA,⁷⁷ glycidyl methacrylate,⁷⁸ or vinyltrimethoxysilane⁷⁹ onto PP indicated that adding styrene as a co-agent successfully enhances grafting yield.^{77,78} (Subsequently, the grafted functional groups can react to form cross-linked PP.^{80–82}) One factor contributing to this improved outcome is the resonance stabilization provided by the styrene phenyl group, which helps to distribute the electron density of the radical and reduces PP's tendency toward β -scission.⁷⁸ Inspired by these studies, our group recently developed a phenylacrylate-based dynamic covalent cross-linker featuring a styrene-like structure.⁷⁵ This dynamic covalent cross-linker demonstrated

enhanced cross-linking efficiency when compared to an analogous methacrylate-based dynamic covalent cross-linker. Additionally, it successfully produced networks with low peroxide content levels and relatively low-MW PP, while the methacrylate-based cross-linker could not.⁷⁵

In addition to styrene, various polyfunctional vinyl monomers have been used as additives to facilitate the linking of two PP macroradicals.⁵⁸ Increasing the functionality of these monomers results in a corresponding increase in gel content, indicating a greater degree of PP cross-linking;⁸³ for example, using only 1 wt% of pentaerythritol tetraallyl ether (a tetrafunctional monomer), as an additive, >80% gel content was achieved for the resulting cross-linked PP.^{58,84} Divinylbenzene (DVB, $C_6H_4(CH=CH_2)_2$) is a difunctional vinyl monomer which corresponds to styrene ($C_6H_5-CH=CH_2$) with an aromatic hydrogen replaced by a second vinyl group. As a result, DVB provides both resonance stabilization and a source of permanent cross-links during radical-based processing.^{64,85} One study⁶⁴ focused on achieving high melt strength by incorporating low levels of DVB (0.5 wt%) during reactive melt extrusion; utilizing ultrasound during reactive extrusion (and with styrene as another additive), high melt strength PP was obtained with a 21% gel content.⁶⁴ Utilizing a PP copolymer with low ethylene content, another study⁸⁵ use of more than 2 wt% DVB as an additive of during reactive melt extrusion led to samples with a loss factor ($\tan \delta$) less than 1 at 200 °C across all measured frequencies, which suggests the formation of a percolated, permanently cross-linked network. Furthermore, like styrene, DVB itself has also been used to improve the grafting efficiency of other vinyl monomers.^{86,87}

Considering these factors, we aimed to use vinyl-containing aromatic compounds, like styrene and DVB, as stabilizing additives during the radical-based cross-linking of PP with a methacrylate-based dynamic cross-linker. However, styrene, with a boiling point of 145 °C,⁸⁸ is highly volatile at the melt processing temperatures of isotactic PP, and thus is incompatible with the batch, open, melt mixer system used in our study. Instead, as shown in Figure 1a, we used 2-vinylnaphthalene (VN, boiling point = 271 °C), which structurally resembles styrene but has an additional aromatic ring and lacks any extraneous units like methyl groups, as an alternative to styrene. We also studied DVB, which serves as both a stabilizing additive and a permanent cross-

linker. For the dynamic covalent cross-linker, we chose bis(4-methacryloyloxyphenyl) disulfide (BiPheS methacrylate or BPMA); BPMA features aromatic disulfide bonds with a catalyst-free dynamic character.⁸⁹ We used dicumyl peroxide as the radical initiator.

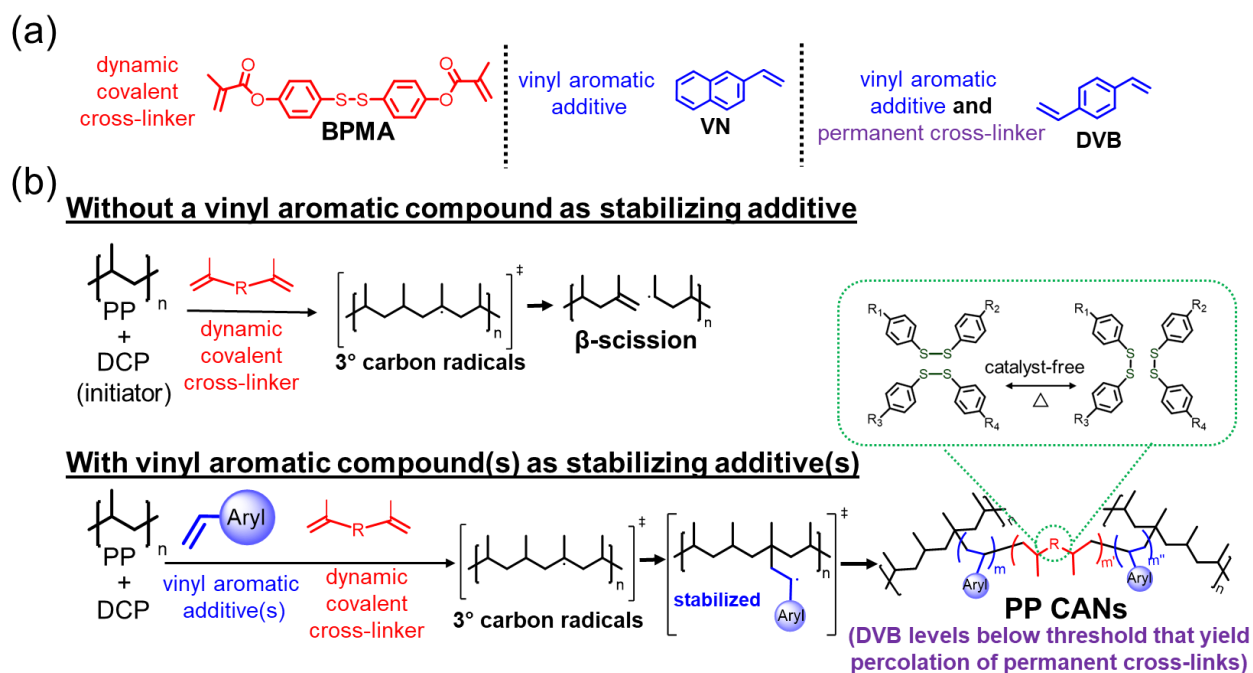


Figure 1. (a) Structures of BPMA, a methacrylate-based dynamic covalent cross-linker; 2-vinylnaphthalene (VN), a vinyl aromatic additive; and divinylbenzene (DVB), a vinyl aromatic additive which also functions as a permanent cross-linker. (b) Dominant reactions during one-step, radical-based reactive processing of PP homopolymer at 180 °C. Using only BPMA and DCP, β -scission is dominant; when VN or DVB (or both) is added as a stabilizing additive for tertiary carbon radicals, dynamic cross-linking becomes dominant. (The amount of DVB must be below the level that yields percolation of permanent cross-links.) See Figures S1 and S2 for additional possible reactions.

During our radical-based reactive processing, PP macroradicals can react with the carbon-carbon double bonds of BPMA (or a chain of polymerized BPMA molecules), thus dynamically

linking the PP chains.¹⁹⁻²¹ Figure 1b (with additional possible side reactions shown in Figures S1 and S2) illustrates how the vinyl aromatic additives suppress β -scission by promoting the formation of stabilized macroradical intermediates;⁷⁸ DVB serves an additional role by enabling the formation of some permanent cross-links.^{64,85} Here, we demonstrated that the use of 2.0 and 4.0 molar equivalents of VN (relative to BPMA) results in PP CANs with robust network character and significant creep suppression. Similarly, 0.5 equivalent of DVB (relative to BPMA) also results in robust PP CANs; however, due to its additional role as a permanent cross-linker, higher levels of DVB lead to the percolation of permanent cross-links and the formation of PP thermosets rather than PP CANs. We additionally demonstrated a complementary effect by combining the two additives: the use of 2.0 molar equivalents of VN and 0.5 molar equivalents of DVB results in a PP CAN with an enhanced network response and improved melt-state creep resistance compared to CANs using a single additive. Notably, all our PP CANs exhibit complete recovery of cross-link density within experimental uncertainty after multiple reprocessing cycles. These results highlight the potential of this method for upcycling virgin or waste thermoplastic PP into PP CANs.

EXPERIMENTAL

Materials. All compounds were used in their as-received forms unless specified otherwise. Bis(4-hydroxyphenyl) disulfide (99.91%) was sourced from Ambeed. Polypropylene (PP, isotactic, melt flow index (MFI) of 12 (12 g/10 min at 230 °C/2.16 kg), Sigma-Aldrich reports M_w ~250,000 g/mol, M_n ~67,000 g/mol), 2-vinylnaphthalene (95%), divinylbenzene (85%), methacryloyl chloride (97%), potassium carbonate (99.0%), magnesium sulfate (99.5%), dicumyl peroxide (98%), o-xylene (98%), decalin (99%) and ethanol (99.5%) were acquired from Sigma-Aldrich. Anhydrous acetonitrile (99.8%), chloroform (99.8%), dichloromethane (99.5%), and methanol (99.8%) were sourced from Fisher Scientific. Acetonitrile was dried using activated 4-Å molecular sieves for no less than 48 h before use.

Synthesis of Bis(4-methacryloyloxyphenyl) Disulfide (BiPheS Methacrylate, BPMA) Cross-linker. Bis(4-methacryloyloxyphenyl) disulfide (BiPheS methacrylate, BPMA) was synthesized

following a literature procedure.⁸⁹ ¹H NMR spectra (500 MHz, CDCl₃): δ 7.56–7.49 (m, 4H), 7.14–7.06 (m, 4H), 6.36 (t, *J* = 1.2 Hz, 2H), 5.78 (p, *J* = 1.5 Hz, 2H), 2.07 (t, *J* = 1.3 Hz, 6H). ESI MS (ESI): *m/z* found [M–H⁺] for [C₂₀H₁₈O₄S₂–NH₄⁺] 404.11 (calcd 404.52).

Chemical Characterization of BPMA. BPMA was analyzed using methods outlined in a previously established procedure.⁸⁹ Proton nuclear magnetic resonance (¹H NMR) spectroscopy was conducted on a Bruker Avance III 500 MHz spectrometer. All spectra were recorded at ambient temperature with deuterated chloroform as the solvent. Electrospray ionization (ESI) mass spectrometry (MS) was performed on a Bruker AmaZon-SL system, integrated with an Agilent 1100 Series high-performance liquid chromatography module, an ESI source, and a 3D ion trap mass analyzer. The measurements used a mobile phase consisting of 20% methanol and 80% dichloromethane, and positive ion spectra were collected.

Radical-Based Reactive Processing of PP. A mixture of PP (powder form, ranging from 0.5 to 2.0 g), 4.0 wt % BPMA and 1.0 wt % DCP (the latter two in powder form, with constant loading throughout this work) was combined with a specified amount of vinyl aromatic additive (VN and/or DVB) in an Atlas Laboratory Mixing Molder, which was pre-flushed twice with neat PP before synthesis. The percentages of BPMA and DCP were calculated based on the mass of PP in the mixer, while the quantity of the vinyl aromatic additive was determined according to specified molar equivalents relative to BPMA. To prepare the sample designated PP/BPMA/2VN, for example, 1.0 g of PP powder was mixed with 40.0 mg of BPMA (0.103 mmol), 10.0 mg of DCP, and 31.9 mg of VN (0.206 mmol, representing 2.00 molar equivalents to BPMA). Two or three steel balls were added to simulate the turbulent mixing conditions typical of reactive extrusion.⁹⁰ The blends were homogenized at 120 rpm for 3 min at 180 °C. While keeping a constant rotational speed, the mixer's rotor was periodically hand-cranked up and down within the mixing chamber to assist the blending process. After blending, the mixtures were removed from the mixer with a spatula and compression-molded at 180 °C using a PHI press (model 0230C-X1) with a 10-ton (~8 MPa) ram force for 15 min, resulting in 0.6-mm-thick 1st mold films.

Reprocessing of Reactively Processed PP Samples via Compression Molding. 1st mold films

were cut into millimeter-sized pieces and compression molded at 180 °C using a PHI press (model 0230C-X1) with a 10-ton (~8 MPa) ram force for 15 min, resulting in 2nd mold films. This procedure was repeated once more, starting with the 2nd mold films to produce 3rd mold films.

Dynamic Mechanical Analysis (DMA). Tension-mode, small-amplitude oscillatory DMA temperature ramps were conducted on samples using a TA Instruments RSA-G2 Solids Analyzer. The storage modulus (E'), loss modulus (E''), and damping ratio ($\tan \delta = E''/E'$) were recorded as functions of temperature. The temperature ramps were performed in airflow at a heating rate of 3 °C/min, a frequency of 1.0 Hz, and an oscillatory strain of 0.03%.

Differential Scanning Calorimetry (DSC). DSC experiments were conducted using a TA Instruments DSC 2500 with 5-mg samples of neat PP and reactively processed PP samples, placed in hermetically sealed aluminum pans under a nitrogen flow. Two heating and cooling cycles were applied to the samples during testing: the samples were heated to 200 °C at a rate of 10 °C/min, maintained at 200 °C for 10 min, then cooled to 0 °C at the same rate, followed by a hold at 0 °C for 10 min. The second heating scan was used to determine the melting peak ($T_{m,peak}$) from the endothermic peak. The percent crystallinity of each sample was calculated by integrating the melting peak from the second heating scan to determine the latent heat of fusion and then dividing this value by 207.1 J/g, the latent heat of fusion for fully crystalline PP.⁹¹

Gel Content Determination. Samples of neat PP and reactively processed 1st mold PP samples (~100 mg, m_0) were immersed in boiling *o*-xylene (or decalin at 180°C) for 24 h, during which the network samples swelled significantly in the solvent. Afterward, the liquid phase was decanted. The samples were then dried under vacuum at 120 °C until no further weight change was observed, yielding the dried gel weight (m_d). Three specimens were measured for each sample, and the gel content percentage was calculated as: gel content % = 100 (m_d/m_0).

Fourier Transform Infrared (FTIR) Spectroscopy. FTIR spectroscopy was performed at room temperature with a Bruker Tensor 37 FTIR spectrophotometer featuring a resolution of 1 cm⁻¹. Spectra were obtained by averaging 16 scans across the range of 4000–600 cm⁻¹.

Creep. Shear-mode creep experiments were conducted by applying a 3.0 kPa stress to sample discs

(0.6 mm thick and 8 mm in diameter) cut from neat PP and 1st mold films of reactively processed PP samples. The tests utilized an Anton-Paar MCR 302e rheometer with an oven-hood attachment and an 8 mm parallel plate fixture. The samples were equilibrated at 180 °C for 5 min with a normal force of 1.0 N. Creep tests were conducted for a duration of 600 s. The viscous creep strain was calculated by extrapolating the slope of the best-fit line for the data between $t = 500$ s and $t = 600$ s for each respective creep curve back to $t = 0$ s and then subtracting this y-intercept from the final strain value at $t = 600$ s.

RESULTS AND DISCUSSION

Synthesis of CANs by Radical-Based Reactive Processing. Efforts to convert thermoplastic PP homopolymer into PP CANs were conducted using melt-state radical-based reactive processing. Neat PP homopolymer, with a melt flow index (MFI) of 12, was ground into powder and reactively processed at 180 °C for 15 min with dicumyl peroxide (DCP) as the radical initiator and BPMA as the dynamic covalent cross-linker, with loadings of 1.0 wt% and 4.0 wt%, respectively, throughout this work. Depending on the sample, varying levels of vinyl aromatic additives (2-vinylnaphthalene and/or divinylbenzene) were also included; see Table 1. The percentages of BPMA and DCP were calculated based on the PP mass in the mixer, and the amount of vinyl aromatic additive was determined according to specified molar equivalents relative to BPMA. Samples were named using the convention PP/BPMA/“molar equivalent of additive to BPMA”“additive type”. For example, the PP/BPMA/2VN sample was created by reactively processing neat PP with DCP, BPMA, and 2.0 molar equivalents of VN relative to BPMA.

Table 1. Formulation, thermomechanical properties, and gel content of neat, thermoplastic PP sample and 1st mold reactively processed PP samples

Sample ^a	Molar equivalent(s) of additive(s) relative to BPMA ^b	E' @ 180 °C (MPa) ^c	Tan δ @ 180 °C ^c	Gel content (%) ^d
PP	0	<0.01	>1	0
PP/BPMA ^e	0	<0.01	>1	13 ± 2
PP/BPMA/1VN	1.0 (VN)	<0.01	>1	27 ± 2
PP/BPMA/2VN	2.0 (VN)	0.12 ± 0.02	0.32 ± 0.02	63 ± 5
PP/BPMA/4VN	4.0 (VN)	0.13 ± 0.02	0.31 ± 0.01	65 ± 5
PP/BPMA/0.5DVB	0.5 (DVB)	0.06 ± 0.01	0.52 ± 0.03	53 ± 3
PP/BPMA/1DVB	1.0 (DVB)	0.13 ± 0.02	0.35 ± 0.02	78 ± 4
PP/BPMA/2DVB	2.0 (DVB)	0.16 ± 0.02	0.20 ± 0.01	88 ± 3
PP/BPMA/2VN/0.5DVB	2.0 (VN) 0.50 (DVB)	0.20 ± 0.03	0.22 ± 0.02	76 ± 3

^aPrecursor has a melt flow index of 12 g/10 min at 230 °C under 2.16 kg load.

^bAll samples (except neat PP) include 4.0 wt% of BPMA and 1.0 wt% of DCP (calculated based on the mass of PP in the mixer); molar equivalents are calculated relative to this quantity of BPMA.

^cDetermined by DMA

^dFrom swelling tests with *o*-xylene; error bars represent one standard deviation of three measurements. Results were further validated by swelling tests using decalin at 180 °C.

^eData for the PP/BPMA sample is taken from ref. 75 (referred to as “PP12/BPMA4-1” in that work).

To determine the amount of VN needed to suppress scission and promote cross-linking, we created initial mold samples with 0, 1, 2, and 4 molar equivalents of VN to BPMA, designated as PP/BPMA, PP/BPMA/1VN, PP/BPMA/2VN, and PP/BPMA/4VN, respectively. Flory’s theory of ideal rubber elasticity indicates that the tensile modulus (E) in the rubbery plateau regime of an ideal rubber is linearly proportional to the cross-link density.⁷³ In the quasi-rubbery plateau regime of our networks, we can approximate $E(T)$ using $E'(T)$. Additionally, a network with cross-links exceeding the gel point should display a loss factor (tan δ) of less than 1.⁹² Therefore, we assessed the robustness of each film by measuring its tensile storage modulus (E') (Figure 2a) and tan δ (E''/E') (Figure S3a) as a function of temperature using DMA. Quantitative values of E' and tan δ at 180 °C are compiled in Table 1. Neither PP/BPMA nor PP/BPMA/1VN demonstrated any

significant network behavior, as indicated by the lack of a quasi-rubbery plateau and $\tan \delta$ greater than 1 in the melt state. (We note that PP/BPMA/1VN exhibited a higher E' value than PP/BPMA throughout the melt transition temperature range, suggesting β -scission could be partially suppressed by the presence of VN.⁹³) In contrast, PP/BPMA/2VN exhibits a pronounced quasi-rubbery plateau with an E' of 0.12 ± 0.02 MPa and a $\tan \delta$ of less than 0.4 at 180 °C, indicating effective network formation. Increasing VN to 4.0 molar equivalents (PP/BPMA/4VN) did not significantly improve the quasi-rubbery plateau response beyond experimental uncertainty ($E' = 0.13 \pm 0.03$ MPa at 180 °C). The gel content data in Table 1 further supports these findings. Gel content progressively increased with higher amounts of VN, starting from an average of 13% for

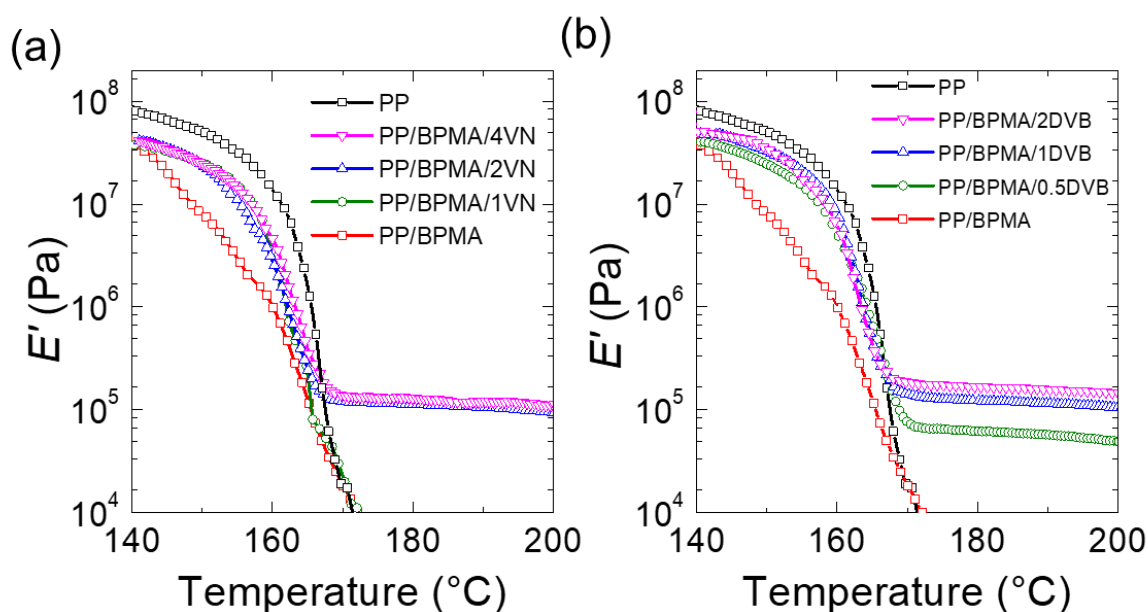


Figure 2. Tensile storage modulus (E') as a function of temperature for neat PP and 1st molds of reactively processed PP samples. Reactively processed PP samples were synthesized at 180 °C using 1.0 wt% DCP and 4.0 wt% BPMA along with (a) 0, 1, 2, or 4 molar equivalents of 2-vinylnaphthalene (VN) or (b) 0, 0.5, 1, or 2 molar equivalents of divinylbenzene (DVB) with respect to BPMA. The sample naming scheme is PP/BPMA/“molar equivalents of additive to BPMA”“additive type”.

PP/BPMA to 27% for PP/BPMA/1VN and reaching 63% for PP/BPMA/2VN. The gel contents of PP/BPMA/2VN and PP/BPMA/4VN are comparable, remaining within the range of experimental uncertainty. These results validate the role of VN in suppressing scission and promoting cross-link formation during radical-based reactive processing. Additionally, these findings indicate that 2.0 equivalents of VN are sufficient to realize approximately fully the benefits of radical stabilization.

In contrast to VN's single carbon-carbon double bond, DVB contains two carbon-carbon double bonds per molecule. As such, DVB plays a dual role by stabilizing tertiary carbon radicals and facilitating the formation of permanent cross-links. To investigate the effect of DVB as an additive, we synthesized a series of reactively processed PP samples with varying amounts of DVB, including 0.5, 1, or 2 molar equivalents relative to BPMA; we designated the resulting materials as PP/BPMA/0.5DVB, PP/BPMA/1DVB, and PP/BPMA/2DVB, respectively. Due to the difference in the number of double bonds in DVB and VN, these DVB-added samples can be considered analogous to the previous VN-added samples PP/BPMA/1VN, PP/BPMA/2VN, and PP/BPMA/4VN, respectively. We conducted DMA on these samples, resulting in E' (Figure 2b) and $\tan \delta$ (Figure S3b) data as a function of temperature. Results at 180 °C are summarized in Table 1. While the addition of 1.0 molar equivalent of VN to BPMA was unable to generate a rubbery plateau response, the addition of 0.5 molar equivalents of DVB to BPMA established percolated cross-links, as evidenced by a quasi-rubbery plateau with $E' = 0.06 \pm 0.01$ MPa at 180 °C and a gel content of 53%. Increasing the DVB concentration to 1.0 or 2.0 molar equivalents relative to BPMA further improved both the gel content and the quasi-rubbery plateau values. Notably, with PP/BPMA/1DVB and PP/BPMA/2DVB, $E'(180\text{ °C})$ increased to 0.13 ± 0.02 MPa and 0.16 ± 0.02 MPa at 180 °C. These results confirm that the use of DVB as an additive also assists in the formation of PP networks with percolated cross-links.⁸⁵

We characterized thermal properties using DSC (Figure S4), with melting transition peak ($T_{m,peak}$) and crystallinity (χ_c) values summarized in Table 2. PP is semicrystalline, and both β -scission and cross-linking disrupt the formation of crystals, which reduces lamella thickness relative to the neat PP precursor,^{94,95} leading to decreases in both $T_{m,peak}$ and χ_c . All reactively

processed samples showed lower $T_{m,peak}$ and χ_c values compared to neat PP. Furthermore, for samples with the same molar equivalent of double bonds provided by the vinyl aromatic additive (e.g., PP/BPMA/2VN and PP/BPMA/1DVB), DVB results in a more significant reduction in χ_c than VN. These findings suggest that DVB facilitates network formation more effectively than VN, which can be attributed to the creation of some permanent cross-links due to the second double bond in the DVB molecule.

Table 2. Thermal properties of neat PP sample and 1st mold reactively processed PP samples

Sample	$T_{m,peak}$ (°C) ^a	χ_c (%) ^b
PP	168 ± 1	45 ± 1
PP/BPMA	161 ± 1	33 ± 2
PP/BPMA/1VN	159 ± 1	39 ± 1
PP/BPMA/2VN	161 ± 1	37 ± 1
PP/BPMA/4VN	161 ± 1	35 ± 1
PP/BPMA/0.5DVB	159 ± 1	33 ± 1
PP/BPMA/1DVB	161 ± 1	34 ± 1
PP/BPMA/2DVB	162 ± 1	34 ± 1
PP/BPMA/2VN/0.5DVB	159 ± 1	36 ± 1

^aDetermined from DSC measurements of the melting peak

^bCalculated as the ratio of the latent heat of fusion of the sample measured from DSC to the latent heat of fusion for 100% crystalline PP (207.1 J/g)⁹¹

Dynamic Character and Reprocessability of Reactively Processed PP Samples. The DMA analysis in the previous section showed that 1st mold samples of PP/BPMA and PP/BPMA/1VN did not display network responses and thus did not have percolated cross-links, whereas PP/BPMA/2VN, PP/BPMA/4VN, PP/BPMA/0.5DVB, PP/BPMA/1DVB, and PP/BPMA/2DVB were networks. Previous research has indicated that BPMA cross-links in CANs formed via free radical polymerization facilitate catalyst-free, reversible exchange reactions, allowing the cross-linked networks to be processed at elevated temperatures.⁸⁹ The exchange mechanism of these

cross-links involves a $[2 + 1]$ radical-mediated process, combining both associative and dissociative characteristics;⁹⁶⁻⁹⁸ in our previous investigation of BPMA-based PE and ethylene-based copolymer CANs produced through reactive processing, the aromatic disulfide bonds displayed significant dissociative behavior as the dominant mechanism.³³

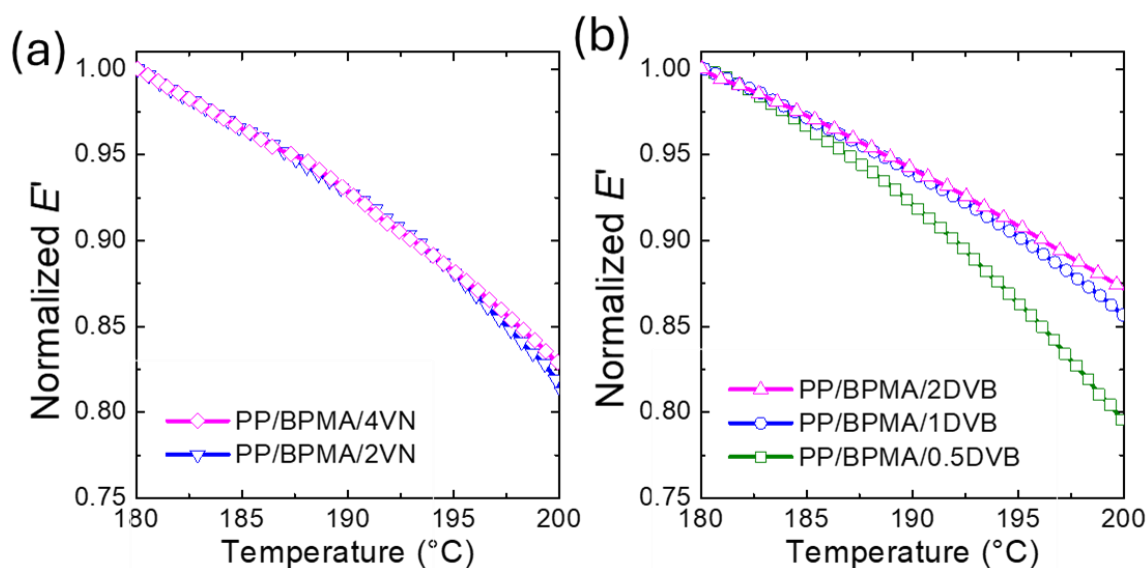


Figure 3. Normalized E' ($E'(T)$ relative to E' at 180 °C) as a function of temperature in the quasi-rubbery plateau regimes of 1st mold PP CANs made with BPMA dynamic covalent cross-linker and (a) VN or (b) DVB. The sample naming scheme is PP/BPMA/“molar equivalents of additive to BPMA”“additive type”.

The loss of network cross-link density with increasing temperature due to dissociative aromatic disulfide chemistry can be quantified using normalized E' ($E'(T)$ relative to E' at 180 °C) as a function of temperature in the quasi-rubbery plateau regime of the five 1st mold cross-linked networks (Figure 3). All samples showed a decrease in E' , and thus the cross-link density, with increasing quasi-rubbery plateau temperature.⁷³ Similar trends in PP/BPMA/2VN and PP/BPMA/4VN indicate that the increased VN content did not significantly modify the dynamic character of the resulting networks. In contrast, compared to PP/BPMA/0.5DVB,

PP/BPMA/1DVB and PP/BPMA/2DVB showed smaller decreases in normalized E' as the temperature increased, indicating that less cross-link density was lost with rising temperature. These results can be attributed to the increased levels of permanent cross-links (which lack dynamic character and, therefore, cannot dissociate) in PP/BPMA/1DVB and PP/BPMA/2DVB. This also makes it clear that DVB levels should be carefully controlled, as too high a fraction of permanent cross-links, which leads to percolated permanent cross-links, will render the networks unable to be reprocessed.⁵⁷

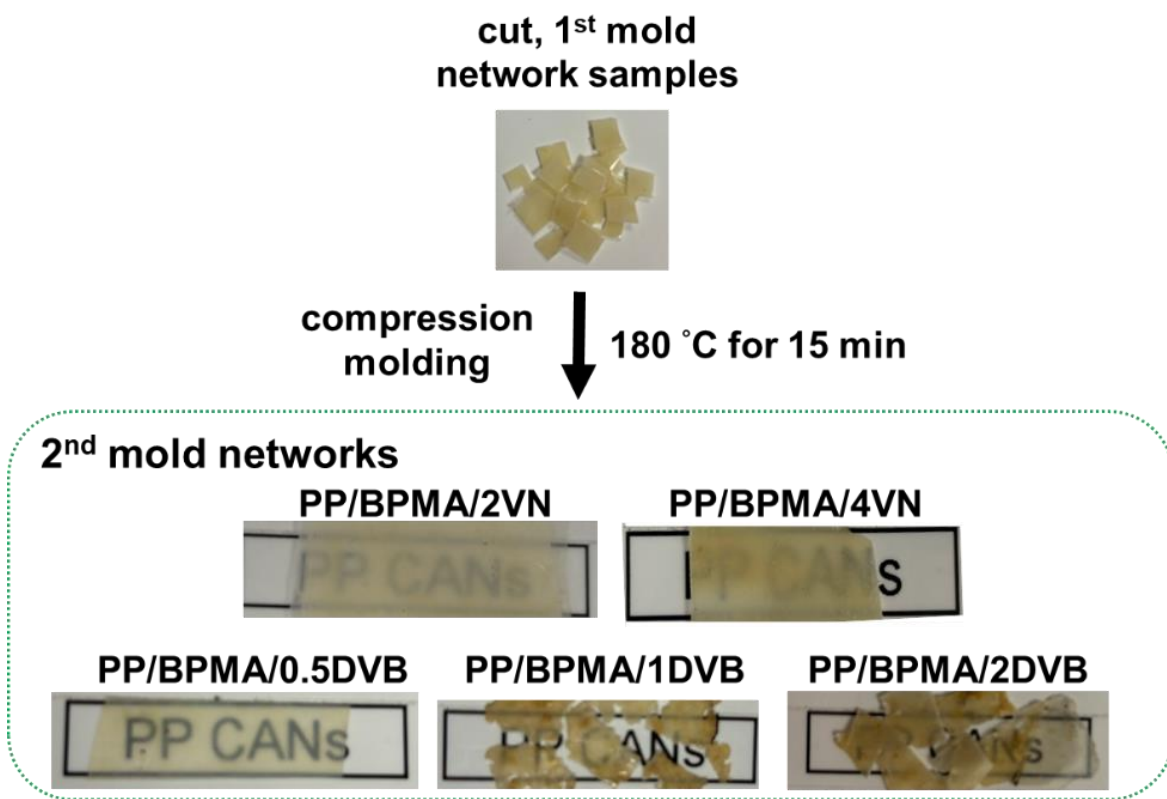


Figure 4. Images of 2nd mold samples of PP/BPMA/2VN, PP/BPMA/4VN, PP/BPMA/0.5DVB, PP/BPMA/1DVB, and PP/BPMA/2DVB reprocessed by compression molding of cut pieces of 1st mold network samples at 180 °C for 15 min; the former three can be molded into robust, consolidated films, whereas the last two cannot be successfully reprocessed (due to percolated permanent cross-links).

Table 3. Remoldability and classification of neat, thermoplastic PP sample and 1st mold reactively processed PP samples

Sample	Remoldable ^a	Sample type
PP	yes	linear polymer
PP/BPMA	yes	non-percolated material ^b
PP/BPMA/1VN	yes	non-percolated material ^b
PP/BPMA/2VN	yes	CAN
PP/BPMA/4VN	yes	CAN
PP/BPMA/0.5DVB	yes	CAN
PP/BPMA/1DVB	no	permanent network
PP/BPMA/2DVB	no	permanent network
PP/BPMA/2VN/0.5DVB	yes	CAN

^aDetermined by whether a robust, consolidated film is produced after compression molding of a cut 1st mold sample at 180 °C for 15 min

^bRefers to samples with liquid-like response above the melting point; the low gel content of such samples (as noted in Table 1) suggests that they largely consist of branched PP, lightly cross-linked PP (lacking a continuous network), unreacted PP chains, and degraded PP chains⁶⁴

Neat PP and reactively processed PP samples that do not lead to network formation (PP/BPMA and PP/BPMA/1VN) can be reprocessed through melt-state processing, as expected. We also aimed to determine which of the five cross-linked networks (PP/BPMA/2VN, PP/BPMA/4VN, PP/BPMA/0.5DVB, PP/BPMA/1DVB, and PP/BPMA/2DVB) were CANs capable of reprocessing or permanent networks that cannot be reprocessed. To evaluate the reprocessability of these five PP networks, the materials were cut into millimeter-sized pieces and compression molded at 180 °C for 15 min. Each network was deemed remoldable if its reprocessed film was healed into a well-consolidated film that appeared homogeneous after this step.^{19-21,33,99} As summarized in Table 3 and Figure 4, the two networks containing VN as an additive (PP/BPMA/2VN and PP/BPMA/4VN) were remoldable. Among the networks with DVB as an additive, only PP/BPMA/0.5DVB showed remoldability. Importantly, PP/BPMA/1DVB and PP/BPMA/2DVB failed to be remolded, although the cross-link densities are within experimental uncertainty the same as those of the reprocessable PP/BPMA/2VN and PP/BPMA/4VN CANs.

These results indicate that the higher concentrations of DVB indeed resulted in the percolation of *permanent* cross-links, thus preventing reprocessing, whereas the two VN-based CANs displayed percolation of dynamic covalent cross-links. In summary, as shown in Table 3, based on the remoldability criterion, three of the five PP networks are CANs: PP/BPMA/2VN, PP/BPMA/4VN, and PP/BPMA/0.5DVB. In contrast, PP/BPMA/1DVB and PP/BPMA/2DVB are permanently cross-linked networks, i.e., thermosets.

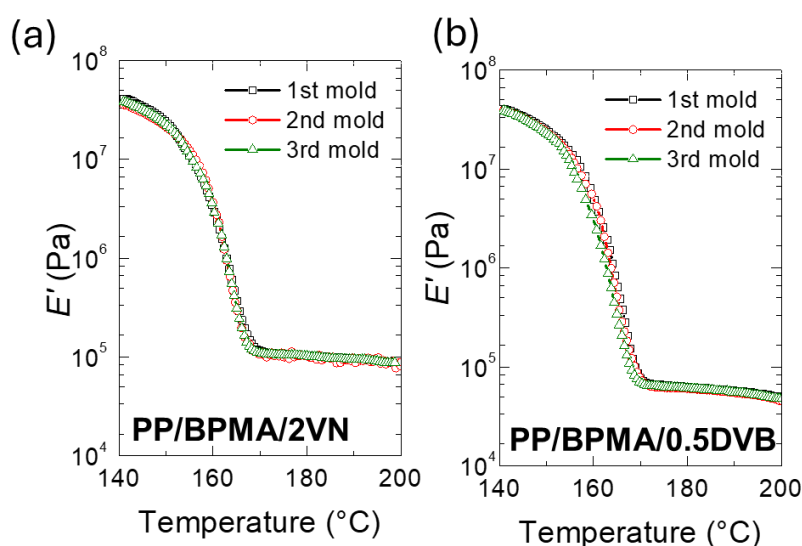


Figure 5. Tensile storage modulus (E') as a function of temperature for 1st, 2nd, and 3rd mold samples of (a) PP/BPMA/2VN or (b) PP/BPMA/0.5DVB; both CANs exhibit full recovery of E' in the rubbery plateau region (and thus, full recovery of cross-link density⁷³ within experimental uncertainty after reprocessing).

For the three PP CANs, DMA was employed to evaluate the recovery of cross-link density in the second and third mold samples. Indeed, according to Flory's ideal rubber elasticity theory,⁷³ the overlapping E' responses from mold to mold (refer to Figure 5 for PP/BPMA/2VN and PP/BPMA/0.5DVB, and Figure S5 for PP/BPMA/4VN; quantitative values at 180 °C are provided in Table S1) indicate a complete recovery of cross-link density after reprocessing for all three

CANs. These results confirm the successful synthesis of robust and fully reprocessable PP CANs. Figures S6 and S7 further verify full property recovery as assessed by DSC and FTIR spectroscopy of the 1st, 2nd, and 3rd mold samples of PP/BPMA/2VN and PP/BPMA/0.5DVB (with DSC parameters summarized in Table S1); the thermograms and spectra exhibit excellent reproduction from mold to mold.

Melt-State Creep Resistance of Reactively Processed PP Samples. In addition to reprocessability, another CAN property of particular importance is creep. At temperatures below the melting point of PP, the ordered crystalline domains of PP are expected to dominate the creep resistance properties.³³ However, melt strength and associated properties are important for foam production; low melt strength polymers can lead to cell coalescence and collapse.¹⁰⁰ Here, we have measured the melt-state creep response at 180 °C, above the PP melt transition endpoint, and thus in the absence of crystallinity.

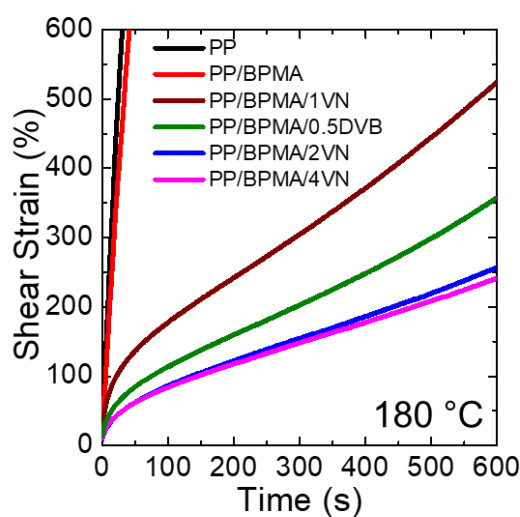


Figure 6. Creep curves for neat PP and 1st molds of reactively processed PP samples with VN or DVB as vinyl aromatic additive. Shear strain (%) was recorded as a function of time under 3.0 kPa of shear stress at 180 °C. Robust dynamic cross-links in the reactively processed PP samples provide creep suppression.

The creep curves (measured with a 3.0 kPa applied shear stress at 180 °C for 600 s) for the CANs are shown in Figure 6 and compared with neat PP and PP/BPMA; the resulting viscous creep strains are summarized in Table S2. Without network formation, neat PP and PP/BPMA display high viscous creep strains of 4700% and 4300%, respectively, typical of flowable thermoplastics. In contrast, the PP/BPMA/2VN, PP/BPMA/4VN, and PP/BPMA/0.5DVB CANs showed more than an order of magnitude reduction in viscous creep strain, with values of 226%, 198%, and 351%, respectively. The similarity of viscous creep strains for PP/BPMA/2VN and PP/BPMA/4VN may be expected as we previously observed little additional benefit in cross-link density from adding VN beyond 2.0 molar equivalents. A greater viscous creep strain was observed for PP/BPMA/0.5DVB compared to the two VN-based CANs, likely due to its lower cross-link density and gel content; in a more loosely cross-linked polymer network, the loss of even a small fraction of cross-links can significantly disrupt the percolation of cross-links, resulting in increased creep.²¹ However, as demonstrated in the previous section, significantly increasing DVB loading to 1.0 or 2.0 molar equivalents during reactive processing is not a viable approach for achieving creep suppression due to permanent network formation, which ultimately undermines the goal of reprocessability. As shown below, we aimed to achieve further enhancement in the properties of our CANs by investigating the use of a blend of VN and DVB additives during reactive processing.

Reprocessability and Melt-State Creep Resistance of CANs using Combined Additives. As previously mentioned, our PP/BPMA/0.5DVB CAN exhibited lower cross-link density than the two VN-based CANs. Our DMA analysis concerning the development of a quasi-rubbery plateau indicated that adding 2.0 molar equivalents of VN to BPMA is sufficient to ensure sufficient resonance stabilization of radicals during the reactive process for network formation. Similarly, our DMA analysis indicated that adding 0.5 molar equivalent of DVB to BPMA during radical-based reactive processing of PP can lead CAN formation; however, significantly higher levels of DVB result in percolated permanent cross-links, rendering the network thermosets impossible to reprocess, even though in one case (PP/BPMA/1DVB) the overall cross-link density is comparable to those of the two CANs made with VN. Here, we examine the use of a combination of additives,

0.5 molar equivalent of DVB and 2.0 molar equivalents of VN (both relative to BPMA), to create reprocessable PP CANs with improved properties.

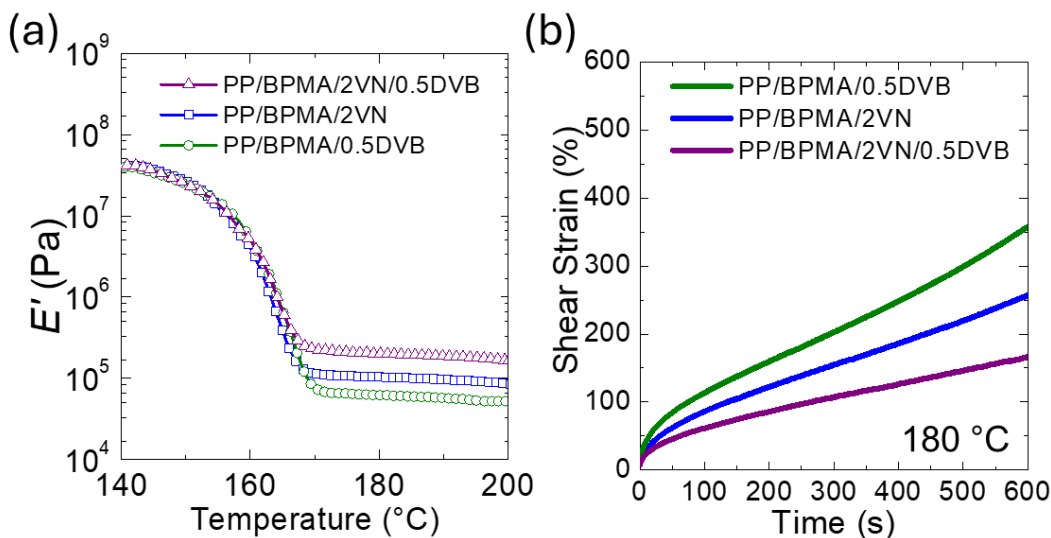


Figure 7. Properties of a 1st mold PP CAN made with both VN and DVB additives (PP/BPMA/2VN/0.5DVB), as compared to samples only made with one or the other: (a) tensile storage modulus (E') as a function of temperature and (b) creep curves with shear strain (%) recorded as a function of time under 3.0 kPa of shear stress at 180 °C. Both E' and creep suppression are further improved by using both additives in tandem.

Following the formulation provided in Table 1, we prepared PP/BPMA/2VN/0.5DVB by incorporating 0.5 molar equivalent of DVB with 2.0 molar equivalents of VN (both with respect to BPMA) during reactive processing. We compared the properties and reprocessability of the resulting network to the counterpart CANs which utilized only a single additive (PP/BPMA/2VN, PP/BPMA/0.5DVB). We measured E' (Figure 7a) and $\tan \delta$ (Figure S8a) as a function of temperature by DMA. From a comparison of E' at 180 °C (refer to Table 1), PP/BPMA/2VN/0.5DVB showed a ~70% increase in cross-link density compared to PP/BPMA/2VN and a ~230% increase as compared to PP/BPMA/0.5DVB. The creep curve of

PP/BPMA/2VN/0.5DVB (measured with a 3.0 kPa applied shear stress at 180 °C for 600 s) is shown in Figure 7b (with the associated viscous creep strain given in Table S2). Due to its higher cross-link density, the viscous creep strain for PP/BPMA/2VN/0.5DVB was only about one-half and one-third of the viscous creep strains for PP/BPMA/2VN and PP/BPMA/0.5DVB, respectively, and a factor of 40 smaller than that of thermoplastic PP. These results suggest that use of a combination of two vinyl aromatic additives, one of which can lead to permanent cross-links, can further enhance cross-link density while retaining melt-state reprocessability and creep resistance.

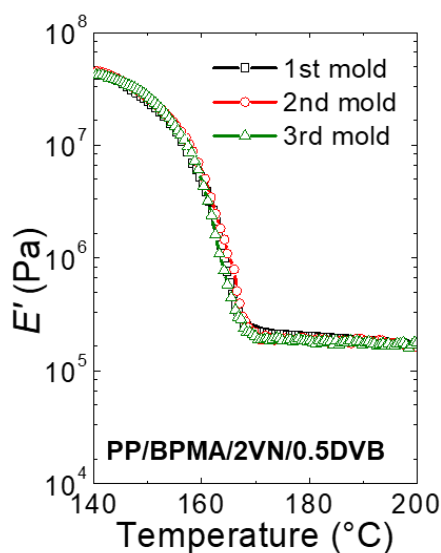


Figure 8. Tensile storage modulus (E') as a function of temperature for 1st, 2nd, and 3rd mold PP/BPMA/2VN/0.5DVB CAN samples, exhibiting full recovery of cross-link density within experimental error after three reprocessing steps.

Figure S8b shows normalized E' ($E'(T)$ relative to $E'(180\text{ °C})$) as a function of temperature in the quasi-rubbery plateau regime for the aforementioned 1st mold CANs; as expected, PP/BPMA/2VN/0.5DVB exhibited a smaller E' loss than both PP/BPMA/2VN and PP/BPMA/0.5DVB. Nevertheless, the level of permanent cross-links in PP/BPMA/2VN/0.5DVB remains below the percolation limit, as a 1st mold sample can be fully reprocessed into a consolidated 2nd mold sample via compression molding at 180 °C for 15 min (Figure S9); this

verifies that PP/BPMA/2VN/0.5DVB can be classified as a CAN. Furthermore, as shown in Figure 8, the excellent overlap of the E' data of the 1st, 2nd, and 3rd mold samples confirm a full recovery of cross-link density in PP/BPMA/2VN/0.5DVB after reprocessing. Figure S10 also presents the DSC and FTIR spectroscopy analyses of the 1st, 2nd, and 3rd mold samples. The full recovery of properties characterized by DSC and FTIR spectral responses further validates the successful synthesis of reprocessable CANs by utilizing a combination of DVB and VN additives to achieve resonance stabilization during the radical-based reactive processing leading to BPMA-based PP CANs formation.

In the future, it will be important to extend this approach by using VN as well as styrene as an additive in the presence and absence of DVB to enhance the cross-linking of PP in the quasi-closed environment of a melt extruder. Additionally, higher boiling point compounds related to styrene should also be tested as additives for use in open mixing units, like the Atlas Laboratory Mixing Molder employed here, and in the quasi-closed environment of a melt extruder.

CONCLUSIONS

We successfully converted PP homopolymer into PP-based CANs through one-step, melt-state radical-based reactive processing. We used constant loadings of DCP (1.0 wt%) as the radical initiator and BPMA (4.0 wt%) as the dynamic cross-linker. However, in the absence of additives, the β -scission of tertiary carbon radicals in PP became the dominant reaction, preventing the generation of networks. We found that vinyl aromatic additives, such as VN or DVB, are essential for stabilizing the tertiary carbon radicals in PP, which facilitates the successful formation of PP networks. The use of 1.0 molar equivalent of VN (relative to BPMA) was able to partially suppress β -scission; however, 2.0 or 4.0 molar equivalents of VN led to robust network formation, as indicated by a quasi-rubbery plateau response measured by DMA and substantial gel contents. The two PP CANs made with 2.0 or 4.0 molar equivalents of VN displayed similar mechanical properties, indicating that the advantages of VN incorporation had reached saturation. The inclusion of DVB as an additive also facilitated the formation of PP networks; however, because

of DVB's role as a permanent cross-linker, this had to be balanced against the risk of permanent cross-linking, which can inhibit reprocessability. Indeed, a network made with 0.5 equivalents of DVB to BPMA was a reprocessable CAN, while networks made with 1.0 and 2.0 equivalents of DVB were not reprocessable. In addition to their robust quasi-rubbery plateau modulus response and reprocessability, the VN-based and DVB-based PP CANs demonstrated strongly suppressed melt-state creep at 180 °C compared to neat PP. We additionally demonstrated that both cross-link density and creep resistance could be further enhanced by incorporating a combination of 2.0 molar equivalents of VN and 0.5 molar equivalent of DVB (relative to BPMA) into a CAN. Finally, we showed that all our PPCANs were reprocessable and fully recovered their cross-link densities and related properties (as confirmed by DMA, DSC, and FTIR spectroscopy) after two successive remolding cycles. Overall, this work demonstrated a new method for preparing robust PP-based CANs through one-step, radical-based reactive processing that utilizes vinyl aromatic additives.

ASSOCIATED CONTENT

Supporting Information

Possible chemistries resulting from radical-based reactive processing, DSC thermograms of neat and reactively processed PP samples, additional DMA and creep data for neat and reactively processed PP samples, DSC thermograms and FTIR spectra of 1st, 2nd, and 3rd mold CANs.

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Notes

A provisional US patent application related to this work has been filed.

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